Musterlösung Serie 20

Diagonalizabiltiy, Cayley-Hamilton

1. Let K be a field, let $n \geqslant 2$, and let

$$A = \begin{pmatrix} 0 & 0 & \cdots & 0 & -c_0 \\ 1 & 0 & \cdots & 0 & -c_1 \\ 0 & 1 & \cdots & 0 & -c_2 \\ 0 & 0 & \cdots & 0 & -c_3 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -c_{n-1} \end{pmatrix} \in M_{n\times n}(K).$$

Prove that

$$\operatorname{char}_A(X) = (-1)^n(X^n + c_{n-1}X^{n-1} + \cdots + c_0).$$

Hint: Use induction.

Solution: First note that for $n = 2$:

$$\det \begin{pmatrix} -X & -c_0 \\ 1 & -X - c_1 \end{pmatrix} = X^2 + c_1 X + c_0 = (-1)^2(X^2 + c_1 X + c_0).$$

Now assume that $n \geqslant 3$ and that we have shown the statement for $n-1$. Expand the determinant with repsect to the first row to obtain

$$\begin{aligned}
&\det \begin{pmatrix} -X & 0 & \cdots & 0 & -c_0 \\ 1 & -X & \cdots & 0 & -c_1 \\ 0 & 1 & \cdots & 0 & -c_2 \\ 0 & 0 & \cdots & 0 & -c_3 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -X - c_{n-1} \end{pmatrix} \\
= \quad & (-X)\det \begin{pmatrix} -X & 0 & \cdots & 0 & -c_1 \\ 1 & -X & \cdots & 0 & -c_2 \\ 0 & 1 & \cdots & 0 & -c_3 \\ 0 & 0 & \cdots & 0 & -c_4 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -X - c_{n-1} \end{pmatrix} + (-1)^{n+1}(-c_0) \\
= \quad & (-X)(-1)^{n-1}(X^{n-1} + c_{n-1}X^{n-2} + \cdots + c_1) + (-1)^n c_0 \\
= \quad & (-1)^n(X^n + c_{n-1}X^{n-1} + \cdots + c_0),
\end{aligned}$$

which concludes the proof.

2. Let A be a $n \times n$-matrix of rank r. Show that the degree of the minimal polynomial of A is smaller or equal than $r + 1$.

   Solution: By the definition of rank, the image of L_A is a subspace of dimension r. Consider the restriction

   $$F := L_A|_{\mathrm{Bild}(L_A)} : \mathrm{Bild}(L_A) \to \mathrm{Bild}(L_A).$$

   The characteristic polynomial $q(X) := \mathrm{char}_F(X)$ of F has degree r, and by Cayley Hamilton, we have $q(F) = 0$.

   For $p(X) := q(X) \cdot X$, we get $v \in K^n$

   $$p(L_A)(v) = (q(L_A) \circ L_A)(v) = q(L_A)(Av) = q(F)(Av) = 0,$$

   and hence $p(L_A) = 0$. Thus, we have $p(A) = 0$. By definition, the minimal polynomial of A divides $p(X)$. Therefore the degree of the first is $\leqslant$ the degree of $p(X)$, hence less or equal $r + 1$.

3. Prove that every 2×2 invertible real matrix belongs to one of the following categories:

   - It is diagonalizable;
   - it is trigonalizable with algebraic multiplicity 2 and geometric multiplicity 1;
   - one can find a basis such that the matrix representation in that basis is

   $$\begin{pmatrix} a & b \\ -b & a \end{pmatrix} \quad \text{with } b \neq 0.$$

   Solution: First note that the characteristic polynomial of any such matrix, denoted p, is a quadratic polynomial with real coefficients. If all of its roots are real, it splits into linear factors in $\mathbb{R}[X]$. In this case, we either have

   - a diagonalizable matrix if the algebraic multiplicity equals the algebraic multiplicity for each root (see the first theorem of the lecture notes eigenvectors.c, in English) or
   - a trigonalizable matrix (see the first theorem on page 4 of the same lecture notes).

   Let us now address the case where the polynomial does not split into linear factors over $\mathbb{R}$. Then, by the fundamental theorem of algebra, it must admit a complex root λ. Moreover, if we write $p(X) = X^2 + c_1X + c_0$, we have that $p(\lambda) = 0$ implies

   $$0 = \overline{p(\lambda)} = \overline{\lambda^2 + c_1\lambda + c_0} = (\overline{\lambda})^2 + c_1\overline{\lambda} + c_0 = p(\overline{\lambda}),$$

   since the coefficients of p are real. Hence the complex conjugate of λ is the other root of p. We are left to show that every such characteristic polynomial arises from a rotation matrix. We prove the following result:

Theorem. *Let A be a 2×2 real matrix with a complex eigenvalue $\lambda \in \mathbb{C}\setminus\mathbb{R}$, and let v be a complex eigenvector corresponding to λ. Then, $A = CBC^{-1}$ for*

$$C = \begin{pmatrix} | & | \\ \Re(v) & \Im(v) \\ | & | \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} \Re(\lambda) & \Im(\lambda) \\ -\Im(\lambda) & \Re(\lambda) \end{pmatrix},$$

where

$$\Re\begin{pmatrix} x+iy \\ z+iw \end{pmatrix} = \begin{pmatrix} x \\ z \end{pmatrix} \quad \text{and} \quad \Im\begin{pmatrix} x+iy \\ z+iw \end{pmatrix} = \begin{pmatrix} y \\ w \end{pmatrix}.$$

In other words, A is similar to a rotation-scaling matrix.

Beweis. We first need to show that the vectors $\Re(v)$ and $\Im(v)$ are linearly independent in order to prove that C is invertible. If not, there exist $x, y \in \mathbb{R}$ such that $x\Re(v) + y\Im(v) = 0$. Then

$$\begin{aligned}(y+ix)v &= y\Re(v) - x\Im(v) + i(x\Re(v) + y\Im(v)) \\ &= y\Re(v) - x\Im(v) \in \mathbb{R}^2.\end{aligned}$$

Hence, on one hand $y\Re(v) - x\Im(v)$ is a complex multiple of v, therefore it is an eigenvector of A with eigenvalue λ. On the other hand, it is a real eigenvector of A, and since A is real, it can only correspond to a real eigenvalue. This is a contradiction.

Let us denote $\lambda = a + ib$ and $v = \left(\begin{smallmatrix} x+iy \\ z+iw \end{smallmatrix}\right)$. Since $\{\Re(v), \Im(v)\}$ forms a basis of $\mathbb{R}^2$,

$$CBC^{-1} = A \iff \begin{cases} A\Re(v) &= CBC^{-1}\Re(v) \\ A\Im(v) &= CBC^{-1}\Im(v) \end{cases}$$

On one hand,

$$\begin{aligned} A\Re(v) + iA\Im(v) &= Av \\ &= \lambda v \\ &= (a+ib)\begin{pmatrix} x+iy \\ z+iw \end{pmatrix} \\ &= \begin{pmatrix} ax-by \\ az-bw \end{pmatrix} + i\begin{pmatrix} bx+ay \\ bz+aw \end{pmatrix}. \end{aligned}$$

On the other hand,

$$Ce_1 = \Re(v) \Leftrightarrow e_1 = C^{-1}\Re(v) \Leftrightarrow CBe_1 = CBC^{-1}\Re(v)$$

and similarly replacing e_1 by e_2 and $\Re(v)$ by $\Im(v)$. Hence

$$CBC^{-1}\Re(v) = CBe_1 = C\begin{pmatrix} a \\ -b \end{pmatrix} = \begin{pmatrix} ax-by \\ az-bw \end{pmatrix} = A\Re(v)$$

and similarly $CBC^{-1}\Im(v) = A\Im(v)$. $\square$

4. Let K be a field, $A \in M_{n\times n}(K)$, and $p \in K[X]$ be a non-trivial polynomial such that $p(A) = 0$. Show that every eigenvalue of A is a root of p.

   Hint: For an eigenvector v of A and a polynomial q over K, state and prove the relationship between v and $q(A) \cdot v$.

   Solution: We proceed as indicated in the hint and consider $q(A)\cdot v$ for a non-trivial polynomial
$$q(X) = b_n X^n + b_{n-1}X^{n-1} + \cdots + b_0 \in K[X]$$
   and an eigenvector v of A. We have
$$\begin{aligned} q(A) \cdot v &= b_n A^n \cdot v + b_{n-1}A^{n-1} \cdot v + \cdots + b_0 I_n \cdot v \\ &= b_n \lambda^n v + b_{n-1}\lambda^{n-1} v + \cdots + b_0 v \\ &= (b_n\lambda^n + b_{n-1}\lambda^{n-1} + \cdots + b_0)v \end{aligned}$$
   Now assume that $p(X) = c_n X^n + c_{n-1}X^{n-1} + \cdots + c_0 \in K[X] \smallsetminus \{0\}$ is such that $p(A) = 0$. We have
$$(c_n\lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_0)v = p(A) \cdot v = 0.$$
   Hence $c_n\lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_0 = 0$, i.e. λ is a root of p.

5. (a) Let A be a $n \times n$-matrix. Prove that the subspace $\langle I_n, A, A^2, \ldots\rangle$ of $M_{n\times n}(K)$ has dimension $\leqslant n$.

   (b) Let $A := \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{pmatrix}$. Find a polynomial $p(X)$ with $p(A) = A^{-1}$.

   Solution:

   (a) *Solution*: The subspace $W := \langle I_n, A, A^2, \ldots, A^{n-1}\rangle$ of $M_{n\times n}(K)$ is generated by n elements and thus has dimension $\leqslant n$. Thus, it is enough to show that $W = \langle I_n, A, A^2, \ldots\rangle$.The inclusion „$\subset$“ is already clear, hence we need to show

   Claim: For all $k \geqslant 0$, we have $A^k \in W$.

   We prove by induction on k.

   Induction start: For $k \leqslant n - 1$ this holds by construction of W.

   Oops: In the case of $n = 0$ this claim is empty, hence no start. But then $M_{n\times n}(K)$ is the zero space and the space in question as well, hence has dimension $n = 0$, as desired. Now let $n \geqslant 1$.

   Induction step: Let $k \geqslant n$.

   Induction hypothesis: The Claim is true for all smaller values of k.

The characteristic polynomial of A is monic of degree n; write it as $\mathrm{char}_A(X) = X^n + \sum_{i=0}^{n-1} a_i X^i$. By Cayley-Hamilton, we have $\mathrm{char}_A(A) = 0$, thus

$$A^n = -\sum_{i=0}^{n-1} a_i A^i.$$

Multiplying with A^{k-n} yields

$$A^k = -\sum_{i=0}^{n-1} a_i A^{i+k-n} = -\sum_{j=k-n}^{k-1} a_{j-k+n} A^j.$$

By induction hypothesis all multiples of A^j on the right side are contained in W; thus A^k also lies in W, which we wanted to show.

(b) The characteristic polynomial of A is

$$\mathrm{char}_A(X) = -X^3 + 6X^2 + 3X - 18$$

By Cayley-Hamilton, we have

$$\mathrm{char}_A(A) = -A^3 + 6A^2 + 3A - 18I_3 = 0,$$

also

$$I_3 = -\frac{1}{18}A^3 + \frac{1}{3}A^2 + \frac{1}{6}A = \left(-\frac{1}{18}A^2 + \frac{1}{3}A^2 + \frac{1}{6}I_3\right) \cdot A.$$

In particular, we get that A is invertible and

$$A^{-1} = -\frac{1}{18}A^2 + \frac{1}{3}A + \frac{1}{6}I_3.$$

6. Prove or disprove: There exists a real $n \times n$-matrix A satisfying

$$A^2 + 2A + 5I_n = 0$$

if and only if n is even.

Solution: The claim is true!

„$\Rightarrow$“: The real polynoial $p(X) := X^2 + 2X + 5$ has the complex zeros $-1 \pm 2i$; these are not real. Now let A be a $n \times n$-matrix with $p(A) = 0$ and n odd. Then the characteristic polynomial of A had odd degree n, and thus has a real zero λ. This is an eigenvalue of A, with corresponding Eigenvector v. For this we have

$$0 = p(A)v = (A^2 + 2A + 5I_n)v = (\lambda^2 + 2\lambda + 5)v = P(\lambda)v,$$

and hence $P(\lambda) = 0$. This is a contradiction to p not having a real zero.

„⇐“: By Cayley-Hamilton, every 2×2-matrix A_2 with characteristic polynomial $p(X)$ satisfies the desired equation, for example the companion matrix $A_2 := \left(\begin{smallmatrix} -2 & 1 \\ -5 & 0 \end{smallmatrix}\right)$. For arbitrary $n \geqslant 0$ let A be the $n \times n$-matrix

$$A := \begin{pmatrix} A_2 & & \\ & \ddots & \\ & & A_2 \end{pmatrix},$$

then we have

$$A^2 + 2A + 5I_n = \begin{pmatrix} A_2 + 2A_2 + 5I_2 & & \\ & \ddots & \\ & & A_2 + 2A_2 + 5I_2 \end{pmatrix} = 0.$$

7. (a) Let f be an endomorphism of a finite-dimensional vector space V, and let $V = V_1 \oplus \ldots \oplus V_r$ with f-invariant subpaces V_i. Show, that the arithmetic, resp. geometric multiplicities of an eigenvalue $\lambda \in K$ of f is equal to the sum of the arithmetic, resp. geometric multiplies of λ as an eigenvalue of the endomorphisms $f|_{V_i}$ of V_i.

   (b) Deduce that f is diagonalizable if and only if $f|_{V_i}$ is diagonalizable for every i.

   (c) Let f and g be endomorphisms for the same finite dimensional vector space V. Show that f and g are *simultaneously diagonalizable* (meaning that there exists a basis of eigenvectors of f which are all also eigenvectors of g) if and only if they commute and are diagonalizable.

   Hint: To prove the the backward direction, first show that each eigenspace of f is g-invariant, i.e. that g maps eigenvectors of f to eigenvectors of f *in the same eigenspace.*

Lösung:

(a) For every $1 \leqslant i \leqslant r$ choose an ordered basis B_i of V_i. Joined in ascending order, these form a basis B of V. The transformation matrix of f with respect to B is then a block diagonal matrix with diagonal blocks $M^{B_i}_{B_i}(f|_{V_i})$ for $1 \leqslant i \leqslant r$. The characteristic polynomial of f thus is the product of the characteristic polynomials of $f|_{V_i}$; i.e.

$$\mathrm{char}_f(X) = \prod_{i=1}^{r} \mathrm{char}_{f|_{V_i}}(X) \tag{1}$$

For every $\lambda \in K$ the arithmetic multiplicity of λ as eigenvalue of f is therefore equal to the sum over $1 \leqslant i \leqslant r$ of the arithmetic multiplicity of λ as eigenvalue of $f|_{V_i}$.

Now consider an arbitrary vector $v = v_1 + \cdots + v_r$ with $v_i \in V_i$. Then we have $f(v) = f(v_1) + \ldots + f(v_r)$ with $f(v_i) \in V_i$ and as $V = V_1 \oplus \ldots \oplus V_r$ is a direct sum, we get

$$f(v_1) + \ldots + f(v_r) = f(v) = \lambda v = \lambda v_1 + \cdots + \lambda v_r$$

if and only if $f(v_i) = \lambda v_i$ for all i.

Important. Here, we really need both the fact that the V_i's are f-invariant *and* that $V = V_1 \oplus \ldots \oplus V_r$ is a direct sum to justify

$$f(v_1 + \cdots + v_r) = f(v) = \lambda v \quad \Leftrightarrow \quad \forall i \in \{1, \ldots, r\} : \ f(v_i) = \lambda v_i.$$

Indeed the backward implication follows directly by factorising by λ but the forward implication is in general false if both conditions are not satisfied.

Hence we have

$$\mathrm{Eig}_\lambda(f) \;=\; \bigoplus_{i=1}^{r} \mathrm{Eig}_\lambda(f|_{V_i})$$

and thus

$$\dim \mathrm{Eig}_\lambda(f) \;=\; \sum_{i=1}^{r} \dim \mathrm{Eig}_\lambda(f|_{V_i}).$$

Therefore, the geometric multiplicity of λ as eigenvalue of f is equal to the sum over $1 \leqslant i \leqslant r$ of the geometric multiplicities of λ as eigenvalue of $f|_{V_i}$.

(b) By a theorem from the lecture an endomorphism of a finite dimensional vector space is diagonalizable if and only if its characteristic polynomial splits into linear factors and dor every of its eigenvalues the geometric multiplicity is equal to the arithmetic multiplicity.

The formula (1) yields that $\mathrm{char}_f(X)$ splits into linear factors if and only if for $\mathrm{char}_{f|_{V_i}}(X)$ does for every i. Consider an arbitrary eigenvalue $\lambda \in K$ of (f). Then we get from (a) and the fact, that the geometric multiplicity is always $\leqslant$ the arithmetic multiplicity, that these multiplicities are equal for f if and only if they are equal for every $f|_{V_i}$. Hence f is diagonalizable if and only if $f|_{V_i}$ is for all i.

(c) Assume that f and g are simultaneously diagonalizable. Then f and g are of course diagonalizable. Let $v_1, \ldots, v_n$ be a basis of V consisting of simultaneous eigenvectors of f and g with eigenvalues $\lambda_1, \ldots, \lambda_n$ for f and $\mu_1, \ldots, \mu_n$ for g. For every element $\sum_{i=1}^n \alpha_i v_i \in V$, we have

$$\begin{aligned} f\left(g\left(\sum_{i=1}^n \alpha_i v_i\right)\right) \;&=\; f\left(\sum_{i=1}^n \alpha_i \mu_i v_i\right) \;=\; \sum_{i=1}^n \alpha_i \mu_i \lambda_i v_i \\ &=\; \sum_{i=1}^n \alpha_i \lambda_i \mu_i v_i \;=\; g\left(\sum_{i=1}^n \alpha_i \lambda_i v_i\right) \;=\; g\left(f\left(\sum_{i=1}^n \alpha_i v_i\right)\right). \end{aligned}$$

Thus f and g commute.

Now assume that f and g commute and are both diagonalizable. As f is diagonalizable, there exist eigenvalues $\lambda_1, \dots, \lambda_r$ of f with $V = \bigoplus_{i=1}^r \mathrm{Eig}_{\lambda_i}(f)$. For every $1 \leqslant i \leqslant r$ and $v \in \mathrm{Eig}_{\lambda_i}(f)$ commutativity of f and g yields:

$$f(g(v)) = g(f(v)) = g(\lambda_i v) = \lambda_i g(v)$$

and thus $g(v) \in \mathrm{Eig}_{\lambda_i}(f)$. The eigenspaces of f are hence g-invariant. As g is also diagonalizable, the map $g|_{\mathrm{Eig}_{\lambda_i}(f)}$ is also diagonalizable for every $1 \leqslant i \leqslant r$ according to (b). Thus there exists a basis B_i of $\mathrm{Eig}_{\lambda_i}(f)$ from eigenvectors of g. Together, we get that $B := B_1 \cup \dots \cup B_r$ is a basis of V consisting of simultaneous eigenvectors of f and g; hence f and g are simultaneously diagonalizable.

8. Let A, B be $n \times n$ matrices over a field F. Prove:

   (a) Prove that if $(I_n - AB)$ is invertible, then $(I_n - BA)$ is invertible, and

   $$(I_n - BA)^{-1} = I_n + B(I_n - AB)^{-1}A$$

   (b) Use part (8a) to show that AB and BA have precisely the same characteristic values in F.

Solution:

(a) Assume that $(I_n - AB)$ is invertible. Then $\det(I_n - AB) \neq 0$ and we have

$$\begin{aligned}
\det(I_n - AB) &= \det\left(\begin{pmatrix} I_n & 0 \\ -A & I_n - AB \end{pmatrix}\right) \\
&= \det\left(\begin{pmatrix} I_n & -B \\ -A & I_n \end{pmatrix}\begin{pmatrix} I_n & B \\ 0 & I_n \end{pmatrix}\right) \\
&= \det\left(\begin{pmatrix} I_n & -B \\ -A & I_n \end{pmatrix}\right)\det\left(\begin{pmatrix} I_n & B \\ 0 & I_n \end{pmatrix}\right) \\
&= \det\left(\begin{pmatrix} I_n & B \\ 0 & I_n \end{pmatrix}\begin{pmatrix} I_n & -B \\ -A & I_n \end{pmatrix}\right) \\
&= \det\left(\begin{pmatrix} I_n - BA & 0 \\ -A & I_n \end{pmatrix}\right) \\
&= \det(I_n - BA)
\end{aligned}$$

so also $\det(I_n - BA) \neq 0$ and $(I_n - BA)$ is invertible as well.

To show the formula

$$(I_n - BA)^{-1} = I_n + B(I_n - AB)^{-1}A$$

Note that this is true if and only if

$$(I_n + B(1 - AB)^{-1}A)(I_n - BA) = I_n$$

and we have

$$\begin{aligned} I_n = (I_n + B(1 - AB)^{-1}A)(I_n - BA) &\iff BA(I_n + B(I_n - AB)^{-1}A) = B(I_n - AB)^{-1}A \\ &\iff BA = B((I_n - AB)^{-1}A(I_n - BA) \\ &\iff I_n = I_n - BA + B(I_n - AB)^{-1}A(I_n - BA) \end{aligned}$$

And if $(I_n - BA)$ is invertible, we can multiply the equation above by $(I_n - BA)^{-1}$, which yields

$$(I_n - BA)^{-1} = I_n + B(I_n - AB)^{-1}A$$

(b) Let $\lambda \in F$. We show that λ is a characteristic value of AB if and only if it is a characteristic value of BA.
First assume $\lambda \neq 0$. Then

$$\lambda I_n - AB = \lambda \left(I_n - \frac{1}{\lambda} AB \right).$$

Hence $\lambda I_n - AB$ is invertible if and only if $I_n - \frac{1}{\lambda}AB$ is invertible. By part (8a), this is equivalent to $I_n - \frac{1}{\lambda}BA$ being invertible. Therefore

$$\lambda I_n - BA = \lambda \left(I_n - \frac{1}{\lambda} BA \right)$$

is invertible if and only if $\lambda I_n - AB$ is invertible. It follows that λ is a characteristic value of AB if and only if it is a characteristic value of BA.
It remains to consider $\lambda = 0$. Note that 0 is a characteristic value of AB if and only if AB is not invertible. This happens if and only if there exists a nonzero vector v such that $ABv = 0$. Then $Bv \neq 0$ (otherwise v would be in $\ker B$ and still give $ABv = 0$), and

$$BA(Bv) = B(ABv) = 0,$$

so BA is not invertible. The converse follows by symmetry. Hence 0 is a characteristic value of AB if and only if it is a characteristic value of BA.
Therefore AB and BA have precisely the same characteristic values in F.

9. Let V be the vector space of all functions $\mathbb{R} \to \mathbb{R}$ which are continuous, i.e. the space of continuous real-valued functions on the real line. Let T be the linear operator on V defined by

$$(Tf)(x) = \int_0^x f(t)dt$$

Prove that T has no characteristic values.

Solution: Let λ be a characteristic value of T, so that $Tf = \lambda f$ for some nonzero function $f \in V$. Then for all $x \in \mathbb{R}$,

$$\int_0^x f(t)\,dt = \lambda f(x).$$

Differentiating both sides with respect to x, we obtain

$$f(x) = \lambda f'(x),$$

hence

$$f'(x) = \frac{1}{\lambda} f(x).$$

Solving this differential equation, we get

$$f(x) = Ce^{x/\lambda}$$

for some constant $C \neq 0$. Substituting into the definition of T, we obtain

$$(Tf)(x) = \int_0^x Ce^{t/\lambda} dt = C \int_0^x e^{t/\lambda} dt = C\lambda \left(e^{x/\lambda} - 1\right).$$

On the other hand,

$$\lambda f(x) = \lambda C e^{x/\lambda}.$$

Hence the equation $Tf = \lambda f$ implies

$$C\lambda \left(e^{x/\lambda} - 1\right) = \lambda C e^{x/\lambda},$$

which simplifies to

$$-C\lambda = 0.$$

Since $C \neq 0$, it follows that $\lambda = 0$. If $\lambda = 0$, then the equation $Tf = 0$ implies

$$\int_0^x f(t)\,dt = 0 \quad \text{for all } x,$$

and differentiating gives $f(x) = 0$ for all x, contradicting that f is nonzero. Therefore T has no characteristic values.